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Noise Avoidance SMOTE in Ensemble Learning for Imbalanced Data

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ABSTRACT Class imbalance is a common problem in many real-world applications. To deal with class imbalance, several techniques, including resampling and ensemble approaches, have been proposed and resampling and ensemble methods have been proven effective for imbalanced data. Moreover, hybrid methods that combine resampling and ensemble have been verified to be highly effective in dealing with imbalance problems. this study proposes new hybrid sampling/ensemble algorithms based on a modification of SMOTE, called NASBoost and NASBagging, which avoids selecting noise samples in the minority class while maintaining diversity among training sets. The proposed sampling method introduces new measures to identify samples that may generate noisy synthetic samples during sampling in SMOTE. Experimental results on 16 imbalanced datasets show that the hybrid of the proposed sampling procedure and ensemble algorithms improves the classification performance by preventing the generation of noise while allowing samples in the minority class to be evenly chosen.

INDEX TERMS Class imbalance, classification complexity, ensemble learning, oversampling, SMOTE.

I. INTRODUCTION

Classification is an important task in pattern recognition. One of the problems that are easily encountered in many real-world applications, such as text categorization [1], intrusion detection [2], [3], and finance [4], [5], is the skewed class distribution, which is known as the class imbalance problem. In this problem, one class (named the minority or positive class) contains a significantly smaller number of samples than other classes (named the majority or negative classes). Problems arising from class imbalance are observed in datasets whose minority class has the highest interest from a learning point of view and entails high costs when misclassified. Class imbalance often causes a bias toward majority classes and a high misclassification rate in a minority class when a classifier is trained using most machine learning algorithms.

Learning from imbalanced data has been intensively studied in the last decade, and several methods, including resampling, cost-sensitive learning, and algorithmic modification, have been proposed to address class imbalance [6], [7]. Resampling is a data-level approach that aims to achieve improved balance among classes in a training dataset [8]. It can be categorized into undersampling and oversampling depending on which class sampling is applied. Undersampling techniques reduce the number of samples in majority classes, whereas oversampling techniques increase the number of samples in a minority class. Meanwhile, cost-sensitive learning imposes higher costs for misclassification in a minority class than on majority classes at the data level, algorithmic level, or a combination of these levels and trains a classifier to minimize the misclassification cost [9]. Algorithmic modification deals with the intrinsic problems of the classification of imbalanced data [10], [11].

Among the three approaches, resampling methods are classifier-independent, and they have been proven simple and effective for imbalanced problems. Another well-known approach for imbalanced data is ensemble learning. In machine learning, an ensemble of classifiers, a set of classifiers whose decisions are combined in some way, is an effective approach for improving the classification performance. This is also true for imbalanced data. There are two main types of ensemble algorithms: boosting and bagging. Boosting trains weak learners sequentially, and a weak learner is trained to correct its predecessor at each

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iteration [12]. Boosting algorithms are effective for imbalanced data, because they impose high weights on misclassified samples at each successive iteration, and the minority class is more likely to be misclassified than the majority classes. Bagging is another category of ensemble algorithms that trains base classifiers using random samples of training data [13], [14].

To enhance the performance of boosting algorithms for imbalanced data, many algorithms, such as SMOTEBoost [15], DATABoost-IM [16], RUSBoost [17], SMOTEbagging [18] and roughly-balanced bagging [19] have been suggested to combine ensemble and sampling. In such methods based on bagging, more effective sampling approaches were introduced instead of random sampling to generate training sets. For example, SMOTEbagging adds synthetic samples to the minority class using SMOTE. On the other hand, sampling techniques for boosting are embedded with a boosting procedure to facilitate learning from imbalanced data. Among the methods with boosting, SMOTEBoost, RAMOBoost, and WOTBoost, which are based on the synthetic minority oversampling technique (SMOTE), which is one of the most popular sampling methods, and DATABoost-IM generates new samples to balance minority and majority classes. The difference between them is the selection of minority class samples to generate synthetic samples. Except for SMOTEBoost, which uses random selection during the sampling procedure, they consider the difficulty level of minority data to generate considerable synthetic samples around the easily misclassified minority class samples.

The hybrid algorithms using SMOTE, however, have a critical drawback in that they do not consider the possibility of introducing noisy positive samples into areas within majority classes, which is also found in other oversampling algorithms based on SMOTE. In addition, hybrid algorithms that assign different sampling probabilities to samples based on difficulty level may lower the diversity between base classifiers by assigning higher weights to samples that are wrong or difficult to correctly estimate in both sampling and learning procedures. Empirically, ensembles tend to achieve enhanced performance when significant diversity exists among individual models [20]–[22].

The main aim of this study is to propose and examine new extensions of SMOTEBoost, named NASBoost (*Noise Avoidance SMOTE*, and SMOTEBagging, named NASBagging, which employ measures for the possibility of generating noisy positive samples in areas within majority classes. Unlike other modifications of SMOTE, the proposed sampling step more precisely identifies noise samples by considering not only the nearest neighbors but also the level of overlap between minority and majority classes around minority class samples. In addition, NAS is designed to avoid generating excessive synthetic samples from the minority class samples that show a high difficulty level in classification or are misclassified in the preceding iteration of the boosting procedure.

The rest of the paper is organized as follows. A brief review of the approaches for addressing class imbalance is indicated in Section II. The proposed method is explained in detail in Section III. The experimental procedures and results are presented in Section IV. The implications and conclusions of this study are provided in Section V.

II. LITERATURE REVIEW

A. CLASSIFICATION FOR IMBALANCED DATA

Several real-world classification problems exist in the web, finance, biology, and telecommunications [7]. Given that a dataset rarely exhibits an equal distribution over all classes, most datasets can be considered imbalanced. Therefore, imbalanced data typically correspond to data showing significant or extreme imbalance in some cases. Class imbalance induces significant compromises in the performance of most standard machine learning algorithms. When typical machine learning algorithms are applied to complex imbalanced datasets, they fail to learn the distributive characteristics of data properly and discriminate minority and majority classes because of the bias toward majority classes, which results in poor accuracy for a minority class.

The problem of learning from imbalanced data has been intensively studied for decades, and several approaches have been developed to address this issue. Methods to learn from imbalanced data are generally categorized into three groups: resampling, cost-sensitive learning, and algorithmic modification [7]. Among them, resampling is the group of methods at the data level that preprocesses a dataset that modifies data distribution to obtain balance between minority and majority classes. Algorithmic level algorithms, including cost-sensitive learning and algorithmic modification, include algorithms that adjust misclassification costs for each class or change decision strategies to impose a bias toward the minority class.

B. RESAMPLING TECHNIQUES FOR IMBALANCED DATA

Resampling techniques are used to obtain better balance among classes to alleviate skew in data distribution. Compared with algorithmic-level approaches for imbalanced data, resampling is versatile because it is independent of learning algorithms [6]. Resampling techniques can be categorized into three groups, as follows.

1) Undersampling: A training set is created by eliminating instances.
2) Oversampling: A training set is created by replicating instances or generating new instances from existing ones.
3) Hybrids: A training set is created using an approach combining undersampling and oversampling

One of the simplest approaches in resampling is to select the majority class samples randomly (random undersampling) to be eliminated or minority class samples to be replicated (random oversampling). However random undersampling can discard useful majority class samples, whereas random oversampling can cause overfitting because

of the substantial exact copies of existing instances [23], [24]. To deal with such problems, advanced methods have been proposed for undersampling and oversampling.

More advanced undersampling techniques are usually based on data-cleaning techniques that remove some instances in the majority classes in the boundary regions between minority and majority classes. The edited nearest neighbor (ENN) removes majority class samples whose majority of k nearest neighbors (kNN) do not agree with their label (k is typically 3) [25]. The condensed nearest neighbor (CNN) selects a set of samples U from a training set X such that 1NN with U can classify the examples almost as accurately as 1NN with X [26]. One-sided selection is an undersampling method that combines CNN and Tomek links, which removes all pairs until all minimally distanced nearest neighbor pairs are of the same class [27], [28]. The neighborhood-cleaning rule (NCR) is a modification of the ENN. NCR eliminates most class samples that are misclassified using kNN or are members of the nearest neighbors of the minority class sample misclassified using kNN [29].

In terms of oversampling, SMOTE is a well-known method to alleviate overfitting. The main idea of SMOTE is to generate new synthetic minority class samples by interpolating among minority class samples that lie together [30]. For interpolation, SMOTE first randomly selects a minority class sample and one of its k minority-class nearest neighbors. A new artificial sample is then created through linear interpolation between the two selected positive samples.

One limitation of SMOTE is that it treats all minority class samples equally, although the difficulty level of samples or the distribution around samples differs. Hence, some modifications of SMOTE have been suggested to provide different weights for resampling or to reflect the different characteristics of the positive samples. The adaptive synthetic sampling approach uses a weighted distribution for different minority class samples in accordance with their difficulty levels in learning and generates additional synthetic samples from the minority class samples with high complexity [31]. Other methods, such as Borderline-SMOTE [32] and Safe-level-SMOTE [33], consider that the classification performance degradation in imbalanced data is affected by not only the level of class imbalance but also other factors related to data distributions [34]. Classification complexity or difficulty varies with the proportion of borderline samples located close to the decision boundary between minority and majority classes or located in the reason where classes overlap and noise samples are located inside the region of the other class [28], [35]. Unlike SMOTE, Borderline-SMOTE and Safe-level-SMOTE generate synthetic samples from borderline samples and exclude synthetic samples from noise samples.

C. ENSEMBLE APPROACH FOR IMBALANCED DATA

Ensemble methods combine several base models to obtain better predictive performance or improve generalizability compared with a single estimator, and they have been proven effective for solving a plethora of machine learning challenges [36]. In recent years, ensembles of classifiers have been considered a possible solution to deal with class imbalance [18], [37], [38]. Two main approaches, namely, bagging and boosting, can be used to obtain ensemble models. In bagging, each base classifier is trained on a random sample of training data [13], [14]. One of the key points for improving the performance of an ensemble is diversity, and sampling is the most common way to incorporate diversity into an ensemble. Bagging uses this strategy. In contrast, boosting iteratively trains weak classifiers through reweighting. After a weak learner is added, weights are modified depending on the accuracy of previous weak learners [14]. Boosting is highly effective in dealing with imbalanced data [39], but bagging may outperform boosting when data contain noise because boosting imposes large weights on noise samples to classify them correctly [40].

Many approaches have been developed using bagging and boosting ensembles to address class imbalance problems. The bagging ensembles for imbalanced data can be categorized as overbagging, underbagging, and under–overbagging depending on the classes to which sampling is applied, similar to resampling [18], [41]. SMOTEBagging combines bagging with SMOTE and oversampling in each iteration; therefore, each training set is completely balanced and consists of different amounts of oversampled minority class and synthetic samples from the minority class by using SMOTE [18]. Roughly balanced bagging combines undersampling with bagging, which has been validated as an effective approach for imbalanced data [19].

In boosting, Adaboost.M2 [42] is usually combined with resampling methods. Typical methods include SMOTEBoost [15], RAMOBoost [43], RUSBoost [17], and EUSBoost [44]. SMOTEBoost and RAMOBoost generate the training sets of base classifiers through SMOTE, a popular oversampling method. RUSBoost and EUSBoost generate training sets of undersampling. Regardless of the resampling steps, these methods reweight instances in accordance with the Adaboost.M2 algorithm. Boosting is effective in dealing with class imbalance, but sequential training of weak learners slows down learning and is more sensitive to noise [45], [46]. Several boosting algorithms, such as SMOTEBoost and RUSBoost, are combined with resampling methods. However, they are more robust than the standard boosting algorithms because the resampling step introduces a high degree of randomness to generate training sets.

III. PROPOSED METHOD

A. MEASURES OF RISK INTRODUCING NOISE SAMPLES BY SMOTE

Different measures of classification complexity, such as feature-based, linearity, and neighborhood measures [47], at data and instance levels are available. The measures of difficulty at the instance level, widely used in sampling methods such as Borderline-SMOTE and Safe-level-SMOTE, are

also measures of classification complexity based on neighborhood. They utilize the number of samples in other classes in the k nearest neighbors (kNN) of samples ($N1$), as follows:

$$N1(i;k) = \frac{\sum_{j \in NN_k(\mathbf{x}_i)} I(y_i \neq y_j)}{k} \tag{1}$$

where $NN_k(\mathbf{x}_i)$ is the set of k nearest neighbors of $\mathbf{x}_i$ and y_i represents the class label of samples $\mathbf{x}_i$, and I is the indicator function, which returns a value of 1 if its argument is true and 0 otherwise. That is, the difficulty level of the samples increases as $N1$ increases. Samples with $N1$ equal to one are typically classified as noisy examples. However, because it only considers kNN, $N1$ cannot reflect the risk of introducing noise samples into areas belonging to the majority classes during the synthesis of new samples of the minority class.

Before introducing new measures of risk introducing noise samples by SMOTE, I want to briefly review how to generate synthetic samples in SMOTE, which is illustrated in Figure 1. SMOTE draws a new sample at a point along the line from the selected positive sample to a positive sample randomly selected from among the k positive nearest samples of the selected sample. In Figure 1, $\mathbf{x}_1$ is an example selected to generate a synthetic sample. When k is 5, $\mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, \mathbf{x}_5$, and $\mathbf{x}_6$ are the five nearest neighbors of $\mathbf{x}_1$ in the positive class. a, b, c, d and e are possible synthetic samples for $\mathbf{x}_1$. In Figure 1, d and e are samples that are likely to be noise, because they are located among majority class samples among small clusters of minority class samples and classify some nearby negative samples as positive. To address this issue, some variants of SMOTE such as Borderline-SMOTE and Safe-level-SMOTE define noise samples in a minority class that cannot be drawn from for sampling. Nevertheless, because noise samples are defined by the proportion of positive samples among the nearest neighbor samples, they can be generated by Borderline-SMOTE and Safe-level-SMOTE. In other words, they do not consider whether nearest neighbors far from the selected positive sample exist among k positive nearest neighbors, which might generate synthetic noise samples. Therefore, it is necessary to define a sample that is likely to generate noise using SMOTE, instead of figuring out noise samples after resampling.

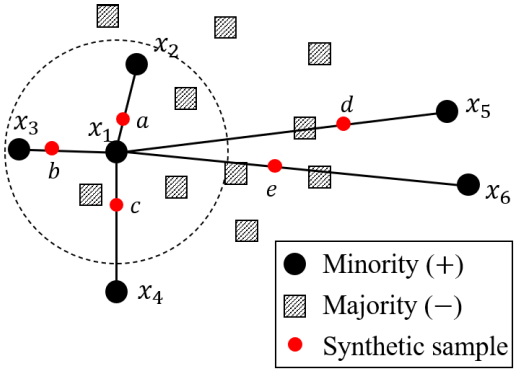


FIGURE 1. Sampling by SMOTE.

This study proposes measures of the possibility to introduce noise samples using SMOTE. The main consideration for these measures is how to measure the overlap between the minority and majority classes.

The first measure is the modification of $N1$ using dynamic local neighborhood. The dynamic local neighborhood of a sample $\mathbf{x}$ with a given k is defined by Eq. 2:

$$NN_d(\mathbf{x};k) = \{\mathbf{x}_i | \parallel \mathbf{x} - \mathbf{x}_i \parallel \leq r, \mathbf{x}_i \neq \mathbf{x}\} \tag{2}$$

where r is the distance to k-th nearest samples in the same class of $\mathbf{x}$.

Based on $NN_d(\mathbf{x};k)$, the first measure of the risk for the i-th sample is defined as the proportion of samples in other classes in the dynamic local neighborhood.

$$R1(i;k) = \frac{\sum_{j \in NN_d(\mathbf{x};k)} I(y_i \neq y_j)}{\parallel NN_d(\mathbf{x};k) \parallel} \tag{3}$$

where $\parallel NN_d(\mathbf{x};k) \parallel$ is the number of samples in the dynamic local neighborhood.

The second measure is the modification of the ratio of the intra/extra-class nearest neighbor distance ($N2$). The ratio of the intra/extra-class nearest neighbor distance computes the ratio of the sum of the distances between individual samples and their nearest neighbors from the same class (intra-class) and the sum of the distances between samples and nearest neighbors from another class (extra-class) [47]:

$$intra_extra = \frac{\sum_{i=1}^{n} \sum_{j \in NN_{1,intra}(\mathbf{x}_i)} d(\mathbf{x}_i, \mathbf{x}_j)}{\sum_{i=1}^{n} \sum_{j \in NN_{1,extra}(\mathbf{x}_i)} d(\mathbf{x}_i, \mathbf{x}_j)} \tag{4}$$

where d is a distance measure between two vectors, $NN_{k,intra}(\mathbf{x}_i)$ and $NN_{k,extra}(\mathbf{x}_i)$ are the set of k nearest neighbors of $\mathbf{x}_i$ from its own class and another class, respectively. $intra_extra$ can take any positive real values; hence, N2, whose value is within a range of 0 to 1, is defined on the basis of $intra_extra$ as follows:

$$N2 = 1 - \frac{1}{1 + intra_extra} = \frac{intra_extra}{1 + intra_extra} \tag{5}$$

Similar to $N1$, $N2$ with $intra_extra$ can also be modified to measure the instance-level difficulty and use kNN as follows:

$$intra_extra_k(i) = \frac{\sum_{j \in NN_{k,intra}(\mathbf{x}_i)} d(\mathbf{x}_i, \mathbf{x}_j)}{\sum_{j \in NN_{k,extra}(\mathbf{x}_i)} d(\mathbf{x}_i, \mathbf{x}_j)} \tag{6}$$

Based on $intra_extra_k(i)$, the second proposed measure, $R2$, is defined as follows:

$$R2(i;k) = \frac{intra_extra_k(i)}{1 + intra_extra_k(i)} \tag{7}$$

In summary, $R1$ can reflect the degree of overlap among classes around each sample, and $R2$ can measure the closeness of samples from another class are to another sample compared with samples from the same class. Therefore, these measures can reflect not only the difficulty level of samples in classification but also whether the synthetic sample using SMOTE is located in the region of the majority class.

B. PROPOSED SAMPLING PROCESS

This study proposes a new sampling method to avoid selecting noise samples from the minority class while maintaining diversity among training sets using $R1$ and $R2$. In terms of generating new samples in SMOTE, high values of $R1$ and $R2$ imply a high possibility of noisy synthetic samples and mean that correct labels are difficult to estimate because of classification complexity. The main role of $R1$ and $R2$ is to avoid selecting minority class samples with a high probability of introducing noise during the sampling procedure. Hence, the proposed algorithm utilizes $R1$ and $R2$ to define risky samples for which SMOTE may generate noisy synthetic samples with high probability. This risk is defined as follows:

$$S_r(i) = \begin{cases} \dfrac{R(i) - R_t}{1 - R_t} & \text{if } R(i) > R_t \\ 0 & \text{if } R(i) \leq R_t \end{cases} \tag{8}$$

where $S_r(i)$ is the risk score of the i-th minority class sample, $R(i)$ represents either $R1(i)$ or $R2(i)$, and R_t is the user-defined threshold value. In other words, no risk of introducing noise samples will occur unless $R(i)$ does not exceed R_t.

To investigate the effectiveness of classification complexity at the instance-level on sampling, NAS is designed such that the difficulty level of samples, as well as the risk score, is considered in the sampling procedure. The proposed modification of SMOTE, NAS, assigns different sampling probabilities to generate a synthetic sample in minority class samples, in accordance with $S_r(i)$ as follows:

$$p(i) = \frac{S_c(i)(1 - S_r(i))}{\sum_{j \in C_+} S_c(j)(1 - S_r(j))}, \ i \in C_+ \tag{9}$$

where C_+ represents the minority (positive) class. The reason that risky samples are not excluded is to promote diversity among weak learners through sampling. In this study, $S_c(i)$ was tested in two different ways. First, $S_c(i)$ is calculated using $N1$. Second, $S_c(i)$ is obtained from the hybridization of $N1$ and the normalized weight distribution of the boosting procedure, $D_t(i, y)$ [15], which can be used only for NASBoost. In the second approach, $S_c(i)$ is updated whenever a weak learner is added, as follows:

$$S_c(i; t) = \frac{\{(T - t - 1)N1(i) + (t - 1)D'_t(i)\}}{T}, \quad t = 1, 2, \ldots, T \tag{10}$$

where T is the number of iterations in boosting, t corresponds to each iteration in the boosting procedure, and $D'_t(i)$ is the score obtained from the normalized weight distribution of the boosting procedure at iteration t. $D'_t(i)$ is defined to be within the range of 0 to 1, the same as $N1(i)$, as follows:

$$D'_t(i) = \frac{D_t(i, y_i)}{\max_{j \in C_+} D_t(i, y_i)}, \quad i \in C_+ \tag{11}$$

Using sampling probability distribution based on Eq. 9, synthetic samples were generated in a manner similar to that used in SMOTE. The sampling procedure is described in Algorithm 1.

Algorithm 1 Synthetic Sample Generation Procedure

Input: p is the sampling probability distribution for the minority class; N is the number of synthetic examples from minority class; k is the number of nearest neighbors.

for $n = 1, \ldots, N$ **do**
  Randomly choose a minority class example, $\mathbf{x}$ from the sampling probability distribution p.
  Find kNN of $\mathbf{x}$ within the minority class, $NN_{k,intra}(\mathbf{x}_i)$.
  Randomly select a minority class sample, $\mathbf{x}_{nn}$ from $NN_{k,intra}(\mathbf{x}_i)$.
  Calculate the difference vector $\boldsymbol{\delta} = \mathbf{x}_{nn} - \mathbf{x}$.
  Create a synthetic sample using the following equation.
  $\mathbf{x}_{syn} = \mathbf{x} + \lambda\boldsymbol{\delta}$ where $\lambda \in [0, 1]$
end for
Output: A temporary training dataset combining the original data with synthetic data

In this study, NAS is combined with boosting and bagging and new ensemble methods with NAS are named NASBoosting and NASBagging, respectively. The boosting procedure of NASBoost is the same as that of other boosting algorithms with oversampling, such as SMOTEBoost, RAMOBoost, and WOTBoost. The bagging procedure for NASBagging is the same as that for SMOTEBagging.

IV. EXPERIMENTS

A. DATA

Several benchmark datasets from various domains are selected for the experiments. Thirteen datasets were obtained from the machine learning repository of the University of California, Irvine [48]. Datasets KC1, and PC1 are software engineering datasets for predicting software defects, obtained from NASA IV&V Facility Metrics Data Program repository.[1] An oil dataset for predicting oil spills from satellite images was obtained from [49]. The multiclass datasets are converted into binary data by assigning the selected label(s) as the positive class (minority class) and all other classes as the negative class (majority class) for comparison. The numerical variables of all datasets are rescaled to have a mean of zero and a standard deviation of one, which prevents some variables with large values from being dominant in the distance calculation where Euclidean distance is used. In addition, the categorical features were discarded. The details of these datasets, such as the number of samples, number of features, and proportion of the selected class to the positive class, are shown in Table 1. The labels of the selected positive classes are enclosed in parentheses under the "Data" column for multiclass datasets.

To validate the effect of noise samples in sampling, this study also modifies the original datasets to create noisy

[1] http://mdp.ivv.nasa.gov/index.html

TABLE 1. Details of data sets.

Data	# of samples	# of attributes	Positive class (%)
Abalone (7)	4,177	7	9.36
Ecoli (imU)	336	8	10.42
Hepatitis	129	17	18.60
Ionosphere	351	34	35.90
Isolet (A,B)	7,797	617	7.70
KC1	2,109	21	15.46
Letter Img (Z)	20,000	16	3.67
Mammography	11,183	6	2.32
Oil	937	48	4.38
Optical digits (8)	5,620	64	9.86
PC1	1,109	21	6.94
Satimage (damp grey soil)	6,435	36	9.73
Spectrometer (≥ 44)	531	93	8.47
US crime (> 0.65)	1,994	100	7.52
Wine quality (≤ 4)	4,898	11	3.74
Yeast (ME2)	1,484	8	3.44

datasets by changing the labels of some majority class samples to positive. To increase noise samples in the minority class, first, the proportion of positive samples in the 20 nearest neighbors for each majority class sample is calculated. Then, as many as 10 % of the minority class size are randomly selected among the majority class samples with a proportion of 0.3 or less. Finally, the labels of the selected majority class samples are changed to positive.

B. EXPERIMENTAL DESIGN

Different models were trained using NASBoost or NASBagging, varying with different combinations of $S_r(i)$ and $S_c(i)$ to investigate the effects of promoting diversity among weak learners and considering the difficulty level of samples with the risk level of introducing noise samples during sampling. For each $R1$ and $R2$, three different classifiers were trained: (1) a classifier considering only $S_r(i)$ ($S_c(i)$ is set to zero for all minority class samples), (2) a classifier considering $S_r(i)$ and static $S_c(i)$ ($S_c(i)$ is equal to either Eq. 1 or 6), and (3) a classifier considering $S_r(i)$ and dynamic $S_c(i)$ ($S_c(i)$ is calculated using Eq. 10). These models are represented as M1, M2, and M3, respectively. M3 is only available for NASBoost. In this study, the Euclidean distance was used to calculate the distance between samples.

Comparison methods are selected among boosting and bagging algorithms combined with SMOTE, such as SMOTEBoost, RAMOBoost, WOTBoost, and SMOTEBagging. This study also tests boosting algorithms with Borderline-SMOTE instead of SMOTE; these are, named B1-SMOTEBoost and B2-SMOTEBoost, two different types of Borderline-SMOTE.

To precisely measure the risk of introducing noise samples by SMOTE, k values of $R1(i)$ and $R2(i)$ are set identical to the number of nearest minority class samples in SMOTE. The threshold value, R_t, was set in the lower quantile of R. The number of nearest minority class samples in SMOTE was set to 5 and the number of nearest neighbors to find dangerous minority instances in Borderline-SMOTE was set at 10. The k of NN_d calculated for $N1$ is set to 5 because the number of nearest minority class samples in SMOTE is 5.

A base classifier selected for boosting and bagging is classification and regression trees (CART).[2] For boosting, the maximum depth of each base classifier is set to 1 to obtain weak learners [50], while an unpruned fully grown tree is used as the base classifier for bagging because unstable base learners are preferred in bagging to increase the diversity among base learners [13]. The number of base classifiers for each ensemble model was set to 50. For comparison, a five-fold cross-validation was applied and repeated 10 times.

In classification problems, the proportion of correctly classified samples over the total number of samples, called the 'accuracy', is a very popular metric for evaluation. However, this metric can be misleading when the data are imbalanced. For imbalanced data, a high accuracy may be obtained even if the classifiers are unable to detect the minority class. Therefore, the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPR) are used to ensure a fair comparison of different classifiers.[3] The receiver operating characteristic (ROC) curve was obtained by plotting the true positive rate against the false positive rate at various threshold settings of the classifiers. Meanwhile, AUROC measures the area under this curve and indicates the probability of a uniformly drawn random positive to be ranked higher than a uniformly drawn random negative. A higher AUROC corresponds to a better performance of the model in separating the positive and negative classes. Similar to ROC, the precision-recall curve is obtained by plotting the precision against the recall; this shows the trade-off between precision and recall for different thresholds. AUPR measures the area under this curve.

C. RESULTS

1) BOOSTING

The different models of NASBoost with the same measures are compared with one another in Tables 2 and 3 for the original datasets. Table 2 summarizes the evaluation results based on AUROC, while Table 3 shows the results obtained by AUPR. These tables provide average values of AUROC and AUPR for each dataset; the standard deviation of AUROC over 10 times cross-validation is enclosed in parentheses. In these tables, the best AUROC values are shown in bold within M1, M2, and M3 for each dataset. "Avg. rank" is the average value of ranks among M1, M2, and M3 for each dataset, and "Win count" refers to the number of datasets for which each model achieved the best AUROC among the three. The win count accounts for the datasets with tie models; thus, the sum of win counts for M1, M2, and M3 may not be the same as the number of datasets, 16.

Regardless of the types of evaluation and risk measures, M2 showed the smallest average rank and the

[2]Owing to limitations of page length, the evaluation results obtained using other types of classifiers are provided in the supplementary material.

[3]Owing to limitations of page length, the evaluation results by F_1 score, which is also widely used metric for imbalanced classification, are provided in the supplementary material.

TABLE 2. Average AUROC values for the original datasets: Comparison between NASBoost models.

Data	$R1$			$R2$		
	$M1$	$M2$	$M3$	$M1$	$M2$	$M3$
Abalone	**0.8514** (0.0027)	0.8510 (0.0028)	0.8487 (0.0033)	**0.8519** (0.0027)	0.8518 (0.0021)	0.8484 (0.0032)
Ecoli	**0.9986** (0.0000)	**0.9986** (0.0000)	0.9984 (0.0000)	**0.9984** (0.0000)	**0.9984** (0.0000)	0.9983 (0.0000)
Hepatitis	0.8315 (0.0021)	**0.8329** (0.0021)	0.8166 (0.0020)	**0.8333** (0.0020)	0.8213 (0.0020)	0.8214 (0.0020)
Ionosphere	0.9471 (0.0283)	**0.9530** (0.0360)	0.9530 (0.0372)	0.9532 (0.0315)	**0.9546** (0.0385)	0.9504 (0.0416)
Isolet	**0.9758** (0.0079)	0.9687 (0.0073)	0.9698 (0.0088)	**0.9758** (0.0076)	0.9674 (0.0088)	0.9709 (0.0092)
KC1	0.7919 (0.0016)	**0.7981** (0.0013)	0.7923 (0.0014)	0.7884 (0.0021)	**0.7933** (0.0021)	0.7890 (0.0025)
Letter Img	0.9823 (0.0000)	0.9929 (0.0000)	**0.9948** (0.0000)	0.9809 (0.0000)	0.9932 (0.0000)	**0.9950** (0.0000)
Mammography	0.9158 (0.0044)	0.9348 (0.0038)	**0.9352** (0.0043)	0.9155 (0.0052)	**0.9362** (0.0041)	0.9321 (0.0057)
Oil	**0.8672** (0.0010)	0.8657 (0.0004)	0.8548 (0.0005)	0.8507 (0.0008)	**0.8721** (0.0004)	0.8553 (0.0005)
Optical digits	0.9822 (0.0024)	**0.9847** (0.0031)	0.9843 (0.0032)	0.9820 (0.0017)	0.9849 (0.0032)	**0.9855** (0.0030)
PC1	0.8080 (0.0150)	**0.8219** (0.0350)	0.8185 (0.0377)	0.8194 (0.0302)	**0.8316** (0.0289)	0.8275 (0.0263)
Satimage	0.9238 (0.0010)	**0.9342** (0.0014)	0.9337 (0.0015)	0.9244 (0.0013)	0.9298 (0.0012)	**0.9313** (0.0012)
Spectrometer	0.9590 (0.0080)	**0.9651** (0.0176)	0.9629 (0.0115)	0.9578 (0.0142)	**0.9664** (0.0130)	0.9643 (0.0098)
US crime	0.9057 (0.0014)	**0.9061** (0.0021)	0.9046 (0.0019)	**0.9040** (0.0013)	0.9000 (0.0023)	0.8923 (0.0012)
Wine quality	0.7970 (0.0000)	**0.8136** (0.0000)	0.8008 (0.0000)	**0.8203** (0.0000)	0.8193 (0.0000)	0.8132 (0.0000)
Yeast	**0.8557** (0.0146)	0.8536 (0.0070)	0.8485 (0.0073)	**0.8459** (0.0168)	0.8457 (0.0088)	0.8404 (0.0110)
Avg. Rank	2.2500	1.4375	2.2500	2.0625	1.6875	2.1875
Win count	5	10	2	7	7	3

TABLE 3. Average AUPR values for the original datasets: Comparison between NASBoost models.

Data	$R1$			$R2$		
	$M1$	$M2$	$M3$	$M1$	$M2$	$M3$
Abalone	0.3087 (0.3087)	**0.3093** (0.3093)	0.2988 (0.2988)	0.3094 (0.3094)	**0.3116** (0.3078)	0.3009 (0.3009)
Ecoli	**0.9824** (1.0000)	**0.9824** (1.0000)	0.9799 (1.0000)	**0.9799** (1.0000)	**0.9799** (1.0000)	0.9774 (1.0000)
Hepatitis	0.6691 (0.9824)	**0.6788** (0.9824)	0.6292 (0.9799)	**0.6668** (0.9799)	0.6375 (0.9755)	0.6418 (0.9774)
Ionosphere	0.9335 (0.6691)	**0.9439** (0.6788)	0.9428 (0.6292)	0.9440 (0.6668)	**0.9500** (0.6293)	0.9413 (0.6418)
Isolet	**0.8427** (0.9335)	0.8054 (0.9439)	0.8129 (0.9428)	**0.8430** (0.9440)	0.7985 (0.9452)	0.8109 (0.9413)
KC1	0.4393 (0.8427)	**0.4516** (0.8054)	0.4366 (0.8129)	0.4372 (0.8430)	**0.4602** (0.7960)	0.4274 (0.8109)
Letter Img	0.8610 (1.0000)	0.9197 (1.0000)	**0.9336** (1.0000)	0.8514 (1.0000)	0.9215 (1.0000)	**0.9350** (1.0000)
Mammography	0.5821 (0.4393)	**0.6531** (0.4516)	0.6407 (0.4366)	0.5784 (0.4372)	**0.6626** (0.4433)	0.6474 (0.4274)
Oil	0.5104 (0.8610)	0.5164 (0.9197)	**0.5169** (0.9336)	0.4987 (0.8514)	**0.5142** (0.9170)	0.4795 (0.9350)
Optical digits	0.9274 (0.5821)	0.9332 (0.6531)	**0.9332** (0.6407)	0.9280 (0.5784)	0.9342 (0.6507)	**0.9359** (0.6474)
PC1	**0.3387** (0.5104)	0.3226 (0.5164)	0.3304 (0.5169)	0.3414 (0.4987)	0.3255 (0.4977)	**0.3433** (0.4795)
Satimage	**0.6664** (0.9274)	0.6283 (0.9332)	0.5849 (0.9332)	**0.6660** (0.9280)	0.5578 (0.9330)	0.5802 (0.9359)
Spectrometer	0.8250 (0.3387)	**0.8309** (0.3226)	0.8113 (0.3304)	0.8296 (0.3414)	0.8395 (0.3196)	**0.8439** (0.3433)
US crime	**0.5723** (0.6664)	0.5691 (0.6283)	0.5382 (0.5849)	**0.5614** (0.6660)	0.5358 (0.5511)	0.5123 (0.5802)
Wine quality	0.2603 (1.0000)	**0.2736** (1.0000)	0.2646 (1.0000)	0.2781 (1.0000)	**0.2947** (1.0000)	0.2701 (1.0000)
Yeast	0.3270 (0.8250)	**0.3400** (0.8309)	0.3355 (0.8113)	0.3282 (0.8296)	**0.3577** (0.8337)	0.3226 (0.8439)
Avg. Rank	2.1250	1.5625	2.2500	1.9375	1.7500	2.2500
Win count	5	9	3	5	8	4

highest win count. Interestingly, M1 showed the slightly better performance than M3. In other words, compared with the case that additionally considers the accuracy of previous classifiers at the sampling step, enhanced results can be obtained by preventing the generation of noise samples.

Next, Tables 3 and 5 summarize the evaluation results for the different models of NASBoost for the noisy datasets.

TABLE 4. Average AUROC values for the noisy datasets: Comparison between NASBoost models.

Data	$R1$			$R2$		
	$M1$	$M2$	$M3$	$M1$	$M2$	$M3$
Abalone	0.8119 (0.0032)	**0.8128** (0.0031)	0.8106 (0.0022)	0.8112 (0.0020)	**0.8119** (0.0022)	0.8051 (0.0031)
Ecoli	**0.9778** (0.0076)	0.9775 (0.0072)	0.9753 (0.0055)	**0.9757** (0.0082)	0.9741 (0.0093)	0.9716 (0.0073)
Hepatitis	0.8044 (0.0202)	**0.8241** (0.0311)	0.8051 (0.0324)	**0.8099** (0.0320)	0.8000 (0.0274)	0.7844 (0.0346)
Ionosphere	0.8927 (0.0147)	**0.8984** (0.0113)	0.8880 (0.0097)	0.8901 (0.0169)	**0.8972** (0.0104)	0.8858 (0.0085)
Isolet	**0.9259** (0.0013)	0.9145 (0.0027)	0.9153 (0.0035)	**0.9212** (0.0027)	0.9018 (0.0037)	0.8962 (0.0024)
KC1	0.7582 (0.0040)	**0.7602** (0.0043)	0.7583 (0.0057)	0.7591 (0.0036)	**0.7614** (0.0040)	0.7533 (0.0068)
Letter Img	0.9312 (0.0014)	0.9420 (0.0009)	**0.9426** (0.0009)	0.9276 (0.0023)	**0.9447** (0.0012)	0.9428 (0.0015)
Mammography	0.8817 (0.0021)	**0.8976** (0.0036)	0.8947 (0.0038)	0.8838 (0.0028)	**0.8979** (0.0030)	0.8918 (0.0033)
Oil	**0.8373** (0.0276)	0.8359 (0.0260)	0.8221 (0.0169)	0.8272 (0.0336)	**0.8375** (0.0244)	0.8316 (0.0294)
Optical digits	0.9381 (0.0012)	0.9398 (0.0019)	**0.9410** (0.0026)	**0.9388** (0.0021)	0.9369 (0.0022)	0.9365 (0.0029)
PC1	0.7848 (0.0174)	**0.7934** (0.0182)	0.7760 (0.0175)	**0.7947** (0.0177)	0.7938 (0.0172)	0.7885 (0.0126)
Satimage	0.8772 (0.0027)	0.8875 (0.0024)	**0.8886** (0.0024)	0.8777 (0.0027)	**0.8852** (0.0026)	0.8826 (0.0025)
Spectrometer	0.9064 (0.0161)	**0.9135** (0.0144)	0.9101 (0.0159)	0.9044 (0.0137)	**0.9074** (0.0206)	0.9025 (0.0161)
US crime	**0.8673** (0.0070)	0.8655 (0.0053)	0.8606 (0.0051)	**0.8613** (0.0075)	0.8563 (0.0059)	0.8452 (0.0062)
Wine quality	0.7671 (0.0087)	**0.7843** (0.0086)	0.7783 (0.0086)	**0.7857** (0.0080)	0.7852 (0.0093)	0.7796 (0.0109)
Yeast	0.8295 (0.0192)	**0.8396** (0.0232)	0.8265 (0.0266)	0.8173 (0.0143)	**0.8208** (0.0224)	0.8180 (0.0176)
Avg. Rank	2.2500	1.5000	2.2500	1.8750	1.4375	2.6875
Win count	4	9	3	7	9	0

TABLE 5. Average AUPR values for the noisy datasets: Comparison between NASBoost models.

Data	$R1$			$R2$		
	$M1$	$M2$	$M3$	$M1$	$M2$	$M3$
Abalone	**0.3027** (0.3027)	0.3024 (0.3008)	0.2998 (0.2998)	**0.3071** (0.3071)	0.3047 (0.3013)	0.3047 (0.3047)
Ecoli	0.9412 (0.9320)	**0.9414** (0.9320)	0.9406 (0.9317)	0.9391 (0.9311)	**0.9403** (0.9318)	0.9350 (0.9320)
Hepatitis	0.6505 (0.9412)	**0.6614** (0.9385)	0.6449 (0.9406)	0.6524 (0.9391)	**0.6659** (0.9366)	0.6216 (0.9350)
Ionosphere	0.8988 (0.6505)	**0.9046** (0.6498)	0.8950 (0.6449)	0.8988 (0.6524)	**0.9083** (0.6177)	**0.9083** (0.6216)
Isolet	**0.7590** (0.8988)	0.7219 (0.9014)	0.7219 (0.8950)	**0.7417** (0.8988)	0.7196 (0.9055)	0.6505 (0.9083)
KC1	**0.4571** (0.7590)	0.4560 (0.7137)	0.4560 (0.7219)	**0.4586** (0.7417)	0.4580 (0.6434)	0.4557 (0.6505)
Letter Img	0.7651 (0.9323)	**0.8308** (0.9294)	0.8106 (0.9323)	0.7393 (0.9323)	**0.8308** (0.9323)	0.8053 (0.9323)
Mammography	0.5696 (0.4571)	**0.6361** (0.4261)	**0.6361** (0.4560)	0.5672 (0.4586)	**0.6409** (0.4519)	0.6294 (0.4557)
Oil	0.4890 (0.7651)	**0.5018** (0.8158)	0.4847 (0.8106)	0.4884 (0.7393)	**0.4945** (0.7761)	**0.4945** (0.8053)
Optical digits	0.8638 (0.5696)	**0.8701** (0.5952)	**0.8701** (0.6361)	0.8651 (0.5672)	**0.8689** (0.6259)	0.8476 (0.6294)
PC1	0.2873 (0.4890)	**0.2991** (0.4904)	0.2802 (0.4847)	**0.3007** (0.4884)	**0.3007** (0.4827)	0.2852 (0.4945)
Satimage	**0.6260** (0.8638)	0.5941 (0.8676)	0.5941 (0.8701)	**0.6234** (0.8651)	0.5733 (0.8536)	0.5177 (0.8476)
Spectrometer	0.7725 (0.2873)	**0.7801** (0.2763)	**0.7801** (0.2802)	0.7643 (0.3007)	**0.7779** (0.2891)	0.7625 (0.2852)
US crime	**0.5326** (0.6260)	0.5215 (0.5630)	0.5208 (0.5941)	**0.5234** (0.6234)	**0.5234** (0.5117)	0.4695 (0.5177)
Wine quality	0.2580 (0.9158)	**0.2722** (0.9302)	0.2694 (0.9285)	0.2846 (0.9158)	**0.2852** (0.9276)	0.2725 (0.9257)
Yeast	0.3315 (0.7725)	**0.3448** (0.7440)	0.3280 (0.7801)	0.3267 (0.7643)	**0.3385** (0.7683)	0.3180 (0.7625)
Avg. Rank	2.0000	1.3125	2.3125	1.8750	1.2500	2.5625
Win count	5	11	3	6	12	2

Similar to the results of the original datasets, M2 showed the smallest average rank and the highest win count. M1 also shows better performance than M3 for the noisy datasets, similar to the results for the original datasets. In addition, M3 with $R2$ for the noisy datasets showed much poorer performance in terms of average rank and win count than it does for the original datasets, which might be due to the sensitivity of the boosting algorithms to the existence of noise

TABLE 6. Average AUROC values for the original datasets: Comparison between NASBoost and the other boosting methods.

Data	NASBoost($R1$)	NASBoost($R2$)	SMOTEBoost	B1-SMOTE	B2-SMOTE	RAMO	WOT
Abalone	0.8510 (0.0028)	0.8518 (0.0021)	**0.8524** (0.0016)	0.8509 (0.0016)	0.8515 (0.0012)	0.8450 (0.0021)	0.8307 (0.0079)
Ecoli	0.9986 (0.0021)	0.9984 (0.0020)	0.9984 (0.0020)	0.9983 (0.0020)	**0.9987** (0.0007)	0.9984 (0.0020)	0.9984 (0.0020)
Hepatitis	**0.8329** (0.0360)	0.8213 (0.0385)	0.8306 (0.0319)	0.7827 (0.0407)	0.7803 (0.0550)	0.7833 (0.0509)	0.7770 (0.0452)
Ionosphere	0.9530 (0.0073)	**0.9546** (0.0088)	0.9442 (0.0057)	0.9526 (0.0084)	0.9530 (0.0078)	0.9454 (0.0123)	0.9427 (0.0105)
Isolet	0.9687 (0.0013)	0.9674 (0.0021)	**0.9759** (0.0019)	0.9571 (0.0028)	0.9548 (0.0042)	0.9734 (0.0023)	0.9714 (0.0025)
KC1	**0.7981** (0.0038)	0.7933 (0.0041)	0.7917 (0.0043)	0.7847 (0.0089)	0.7830 (0.0065)	0.7835 (0.0067)	0.7582 (0.0046)
Letter Img	0.9929 (0.0004)	0.9932 (0.0004)	0.9834 (0.0010)	0.9754 (0.0029)	0.9685 (0.0033)	**0.9970** (0.0003)	0.9950 (0.0011)
Mammography	0.9348 (0.0031)	**0.9362** (0.0032)	0.9175 (0.0031)	0.9264 (0.0033)	0.9311 (0.0028)	0.9336 (0.0023)	0.9253 (0.0047)
Oil	0.8657 (0.0350)	0.8721 (0.0289)	0.8572 (0.0237)	0.8574 (0.0329)	0.8640 (0.0178)	0.8756 (0.0225)	**0.8785** (0.0332)
Optical digits	0.9847 (0.0014)	**0.9849** (0.0012)	0.9824 (0.0011)	0.9789 (0.0020)	0.9760 (0.0018)	0.9840 (0.0015)	0.9842 (0.0019)
PC1	0.8219 (0.0176)	0.8316 (0.0130)	0.8180 (0.0241)	0.8150 (0.0113)	0.8135 (0.0186)	**0.8324** (0.0103)	0.8323 (0.0162)
Satimage	**0.9342** (0.0021)	0.9298 (0.0023)	0.9271 (0.0018)	0.9052 (0.0033)	0.9048 (0.0047)	0.9277 (0.0022)	0.9162 (0.0041)
Spectrometer	0.9651 (0.0070)	**0.9664** (0.0088)	0.9653 (0.0063)	0.9654 (0.0089)	0.9602 (0.0074)	0.9663 (0.0069)	0.9643 (0.0096)
US crime	**0.9061** (0.0057)	0.9000 (0.0046)	0.9037 (0.0036)	0.8924 (0.0053)	0.8948 (0.0058)	0.8755 (0.0091)	0.8672 (0.0092)
Wine quality	0.8136 (0.0084)	0.8193 (0.0046)	0.8197 (0.0065)	0.8135 (0.0072)	**0.8238** (0.0056)	0.8063 (0.0091)	0.7952 (0.0105)
Yeast	**0.8536** (0.0172)	0.8457 (0.0219)	0.8370 (0.0212)	0.8445 (0.0121)	0.8536 (0.0202)	0.8443 (0.0199)	0.8433 (0.0171)
Avg. Rank	2.6250	2.4375	4.0000	5.1875	4.8125	3.6250	4.9375
Win count	5	4	2	0	2	2	1

TABLE 7. Average AUPR values for the original datasets: Comparison between NASBoost and the other boosting methods.

Data	NASBoost($R1$)	NASBoost($R2$)	SMOTEBoost	B1-SMOTE	B2-SMOTE	RAMO	WOT
Abalone	0.3093 (0.3093)	**0.3116** (0.3116)	0.3111 (0.0077)	0.3015 (0.0073)	0.3027 (0.0067)	0.2913 (0.0065)	0.2718 (0.0128)
Ecoli	0.9824 (0.9824)	0.9799 (0.9799)	0.9799 (0.0206)	0.9774 (0.0194)	**0.9843** (0.0106)	0.9799 (0.0206)	0.9799 (0.0206)
Hepatitis	**0.6788** (0.6788)	0.6375 (0.6375)	0.6580 (0.0507)	0.5530 (0.0526)	0.5820 (0.0687)	0.5862 (0.0628)	0.5758 (0.0453)
Ionosphere	0.9439 (0.9439)	**0.9500** (0.9500)	0.9420 (0.0073)	0.9425 (0.0082)	0.9397 (0.0115)	0.9357 (0.0101)	0.9336 (0.0099)
Isolet	0.8054 (0.8054)	0.7985 (0.7985)	**0.8429** (0.0088)	0.7444 (0.0155)	0.7023 (0.0141)	0.8070 (0.0082)	0.7981 (0.0136)
KC1	0.4516 (0.4516)	**0.4602** (0.4602)	0.4394 (0.0098)	0.3958 (0.0159)	0.3974 (0.0156)	0.3976 (0.0144)	0.3488 (0.0081)
Letter Img	0.9197 (0.9197)	0.9215 (0.9215)	0.8649 (0.0040)	0.6665 (0.0251)	0.5891 (0.0290)	**0.9521** (0.0025)	0.9153 (0.0187)
Mammography	0.6531 (0.6531)	**0.6626** (0.6626)	0.5858 (0.0097)	0.5297 (0.0178)	0.5467 (0.0199)	0.6226 (0.0058)	0.6006 (0.0176)
Oil	0.5164 (0.5164)	0.5142 (0.5142)	0.5040 (0.0286)	0.5002 (0.0258)	**0.5364** (0.0317)	0.4622 (0.0335)	0.4710 (0.0563)
Optical digits	0.9332 (0.9332)	**0.9342** (0.9342)	0.9298 (0.0032)	0.9139 (0.0052)	0.8965 (0.0079)	0.9288 (0.0048)	0.9297 (0.0054)
PC1	0.3226 (0.3226)	0.3255 (0.3255)	**0.3449** (0.0438)	0.3080 (0.0307)	0.3017 (0.0195)	0.3378 (0.0144)	0.3262 (0.0267)
Satimage	0.6283 (0.6283)	0.5578 (0.5578)	**0.6683** (0.0067)	0.4106 (0.0122)	0.4082 (0.0157)	0.5645 (0.0130)	0.5022 (0.0174)
Spectrometer	0.8309 (0.8309)	0.8395 (0.8395)	0.8390 (0.0244)	0.8191 (0.0382)	0.7800 (0.0319)	0.8571 (0.0276)	**0.8581** (0.0261)
US crime	**0.5691** (0.5691)	0.5358 (0.5358)	0.5638 (0.0136)	0.4696 (0.0258)	0.4749 (0.0177)	0.3989 (0.0270)	0.3897 (0.0206)
Wine quality	0.2736 (0.2736)	**0.2947** (0.2947)	0.2788 (0.0125)	0.2747 (0.0129)	0.2811 (0.0151)	0.2438 (0.0116)	0.2188 (0.0217)
Yeast	0.3400 (0.3400)	**0.3577** (0.3577)	0.3358 (0.0421)	0.3364 (0.0353)	0.3286 (0.0279)	0.3261 (0.0328)	0.3111 (0.0314)
Avg. Rank	2.6250	2.2500	2.9375	5.5000	5.0000	4.1250	5.1875
Win count	2	7	3	0	2	1	1

samples. In a situation in which the boosting procedure might impose large weights on noise samples, overfitting on the noise samples of the minority class could easily occur even in sampling if the noise samples were frequently selected.

Tables 6 and 7 compare the M2 models of NASBoost with the other boosting algorithms using AUROC and AUPR on the original datasets. For simplicity, except for SMOTEBoost and NASBoost, which also employ the bagging method,

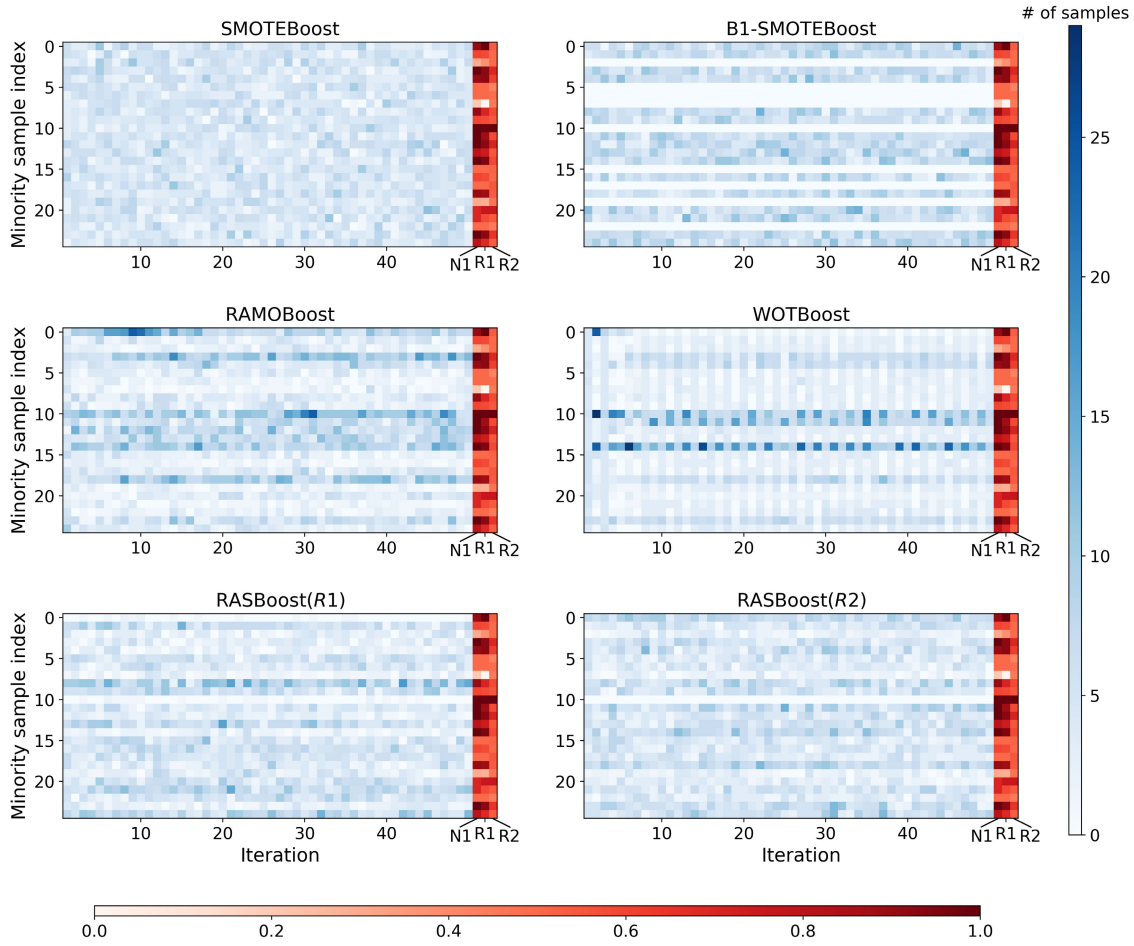


FIGURE 2. Sampling frequency by samples for KC1.

the remaining boosting algorithms with different resampling approaches are denoted by the names of the resampling methods in the tables. In these tables, all seven models were considered together in the calculation of the average ranks and win counts. In terms of the average rank and win count within the seven models, NASBoost shows superior performance to the other methods regardless of the risk measures. The difference in performance between $R1$ and $R2$ is not large, but the difference between NASBoost and the comparison methods seems to be significant. Among the comparison methods, RAMOBoost shows the best performance for AUROC, but SMOTEBoost is the best method for AUPR.

Although NASBoost generally shows good performance, it provides slightly worse results for some datasets, such as those for Oil and Wine quality than the comparison methods. To determine the reason for the performance difference of NASBoost according to the datasets, distributions of $N1$ values were examined by for each dataset. Table 8 shows the proportions of samples in the minority class that satisfy the specific conditions as they vary with the datasets. In the calculation of $N1$, the number of nearest neighbors, k is set at 5 and two different versions of $N1$ values are calculated

TABLE 8. The proportion of the samples in the minority class with high $N1$ ($N1 > 0.9$) and low $N1$ ($N1 < 0.5$).

Data	$N1$ with NN		$N1$ with NN_d	
	High	Low	High	Low
Abalone	0.2583	0.1304	0.1151	0.0895
Ecoli	0.0571	0.8000	0.0286	0.8286
Hepatitis	0.2917	0.4583	0.1667	0.5417
Ionosphere	0.1270	0.6190	0.2222	0.5635
Isolet	0.0117	0.8033	0.0083	0.7933
KC1	0.3006	0.2699	0.1656	0.1933
Letter Img	0.0054	0.9673	0.0068	0.9537
Mammography	0.2077	0.5462	0.2038	0.5038
Oil	0.3171	0.2195	0.4390	0.1707
Optical digits	0.0126	0.9495	0.0054	0.9440
PC1	0.3117	0.2078	0.1558	0.1948
Satimage	0.0687	0.6869	0.0383	0.6470
Spectrometer	0.1333	0.6667	0.0444	0.6000
US crime	0.2667	0.2800	0.1667	0.2467
Wine quality	0.4536	0.0765	0.6175	0.0219
Yeast	0.3922	0.1765	0.3529	0.0588

depending on whether NN or NN_d is used. In this table, 'High' and 'Low' indicate the ratios of the samples in the minority class whose $N1$ values are greater than 0.9 and less than 0.5, respectively. If the value of $N1$ of a sample is greater than 0.9, a sample may be noise. On the other hand, a sample can be considered safe if the value of $N1$ of the sample is less than 0.5. In the case of Wine quality, $N1$ values obtained

TABLE 9. Average AUROC values for the noisy datasets: Comparison between NASBoost and the other boosting methods.

Data	NASBoost($R1$)	NASBoost($R2$)	SMOTEBoost	B1-SMOTE	B2-SMOTE	RAMO	WOT
Abalone	**0.8128** (0.0031)	0.8119 (0.0022)	0.8116 (0.0014)	0.8125 (0.0030)	0.8104 (0.0018)	0.8028 (0.0019)	0.7881 (0.0109)
Ecoli	**0.9775** (0.0072)	0.9741 (0.0093)	0.9768 (0.0083)	0.9616 (0.0115)	0.9439 (0.0120)	0.9602 (0.0079)	0.9640 (0.0101)
Hepatitis	**0.8241** (0.0311)	0.8000 (0.0274)	0.8136 (0.0323)	0.7715 (0.0228)	0.7665 (0.0389)	0.7783 (0.0252)	0.7822 (0.0230)
Ionosphere	**0.8984** (0.0113)	0.8972 (0.0104)	0.8910 (0.0150)	0.8918 (0.0119)	0.8906 (0.0087)	0.8729 (0.0109)	0.8765 (0.0122)
Isolet	0.9145 (0.0027)	0.9018 (0.0037)	**0.9201** (0.0021)	0.8689 (0.0043)	0.8490 (0.0054)	0.8855 (0.0052)	0.8810 (0.0071)
KC1	0.7602 (0.0043)	**0.7614** (0.0040)	0.7593 (0.0048)	0.7515 (0.0075)	0.7516 (0.0055)	0.7458 (0.0062)	0.7235 (0.0095)
Letter Img	0.9420 (0.0009)	**0.9447** (0.0012)	0.9293 (0.0010)	0.9177 (0.0048)	0.9058 (0.0074)	0.9376 (0.0015)	0.9245 (0.0042)
Mammography	0.8976 (0.0036)	**0.8979** (0.0030)	0.8847 (0.0016)	0.8841 (0.0040)	0.8894 (0.0044)	0.8956 (0.0050)	0.8889 (0.0050)
Oil	0.8359 (0.0260)	0.8375 (0.0244)	0.8309 (0.0321)	0.8368 (0.0247)	0.8235 (0.0419)	0.8311 (0.0552)	**0.8425** (0.0427)
Optical digits	**0.9398** (0.0019)	0.9369 (0.0022)	0.9386 (0.0024)	0.9243 (0.0024)	0.9160 (0.0027)	0.9288 (0.0017)	0.9285 (0.0024)
PC1	0.7934 (0.0182)	0.7938 (0.0172)	**0.7947** (0.0137)	0.7798 (0.0107)	0.7829 (0.0083)	0.7933 (0.0121)	0.7814 (0.0162)
Satimage	**0.8875** (0.0024)	0.8852 (0.0026)	0.8792 (0.0028)	0.8605 (0.0032)	0.8578 (0.0026)	0.8748 (0.0029)	0.8557 (0.0043)
Spectrometer	**0.9135** (0.0144)	0.9074 (0.0206)	0.9016 (0.0081)	0.8980 (0.0143)	0.9016 (0.0215)	0.9020 (0.0168)	0.8986 (0.0168)
US crime	**0.8655** (0.0053)	0.8563 (0.0059)	0.8594 (0.0074)	0.8481 (0.0081)	0.8466 (0.0077)	0.8254 (0.0085)	0.8121 (0.0072)
Wine quality	0.7843 (0.0086)	0.7852 (0.0093)	0.7851 (0.0078)	**0.7859** (0.0049)	0.7857 (0.0043)	0.7757 (0.0129)	0.7631 (0.0101)
Yeast	**0.8396** (0.0232)	0.8208 (0.0224)	0.8213 (0.0196)	0.8183 (0.0164)	0.8183 (0.0205)	0.8089 (0.0196)	0.8080 (0.0206)
Avg. Rank	1.8125	2.3125	3.1250	4.8125	5.5000	4.8750	5.5625
Win count	9	3	2	1	0	0	1

TABLE 10. Average AUPR values for the noisy datasets: Comparison between NASBoost and the other boosting methods.

Data	NASBoost($R1$)	NASBoost($R2$)	SMOTEBoost	B1-SMOTE	B2-SMOTE	RAMO	WOT
Abalone	0.3024 (0.3008)	**0.3047** (0.3013)	0.3011 (0.0071)	0.2921 (0.0066)	0.2933 (0.0061)	0.2831 (0.0061)	0.2655 (0.0111)
Ecoli	**0.9414** (0.9320)	0.9403 (0.9318)	0.9226 (0.0020)	0.9184 (0.0020)	0.9318 (0.0020)	0.9230 (0.0020)	0.9233 (0.0020)
Hepatitis	0.6614 (0.9385)	**0.6659** (0.9366)	0.6564 (0.0203)	0.5731 (0.0198)	0.5842 (0.0114)	0.5951 (0.0206)	0.5890 (0.0207)
Ionosphere	0.9046 (0.6498)	0.9083 (0.6177)	0.9119 (0.0497)	**0.9137** (0.0452)	0.9109 (0.0591)	0.9067 (0.0582)	0.9062 (0.0421)
Isolet	0.7219 (0.9014)	0.7196 (0.9055)	**0.7855** (0.0079)	0.6950 (0.0086)	0.6571 (0.0111)	0.7530 (0.0102)	0.7452 (0.0113)
KC1	0.4560 (0.7137)	**0.4580** (0.6434)	0.4271 (0.0074)	0.3898 (0.0140)	0.3912 (0.0133)	0.3909 (0.0078)	0.3489 (0.0130)
Letter Img	0.8308 (0.9294)	0.8308 (0.9323)	0.7961 (0.0001)	0.6182 (0.0001)	0.5477 (0.0001)	**0.8776** (0.0001)	0.8440 (0.0001)
Mammography	0.6361 (0.4261)	**0.6409** (0.4519)	0.5433 (0.0090)	0.4898 (0.0142)	0.5057 (0.0138)	0.5736 (0.0135)	0.5541 (0.0075)
Oil	0.5018 (0.8158)	0.4945 (0.7761)	0.4848 (0.0034)	0.4808 (0.0224)	**0.5138** (0.0262)	0.4383 (0.0023)	0.4445 (0.0170)
Optical digits	**0.8701** (0.5952)	0.8689 (0.6259)	0.8684 (0.0096)	0.8537 (0.0162)	0.8387 (0.0179)	0.8664 (0.0058)	0.8673 (0.0166)
PC1	0.2991 (0.4904)	0.3007 (0.4827)	**0.3369** (0.0309)	0.3024 (0.0306)	0.2984 (0.0317)	0.3315 (0.0316)	0.3201 (0.0530)
Satimage	0.5941 (0.8676)	0.5733 (0.8536)	**0.6307** (0.0031)	0.3955 (0.0050)	0.3930 (0.0067)	0.5385 (0.0041)	0.4818 (0.0052)
Spectrometer	0.7801 (0.2763)	0.7779 (0.2891)	0.7979 (0.0398)	0.7798 (0.0306)	0.7442 (0.0179)	0.8177 (0.0139)	**0.8210** (0.0257)
US crime	0.5215 (0.5630)	0.5234 (0.5117)	**0.5306** (0.0061)	0.4465 (0.0108)	0.4525 (0.0142)	0.3828 (0.0118)	0.3745 (0.0171)
Wine quality	0.2722 (0.9302)	**0.2852** (0.9276)	0.2651 (0.0005)	0.2617 (0.0005)	0.2677 (0.0005)	0.2328 (0.0005)	0.2116 (0.0005)
Yeast	**0.3448** (0.7440)	0.3385 (0.7683)	0.3186 (0.0249)	0.3186 (0.0352)	0.3108 (0.0289)	0.3089 (0.0237)	0.2948 (0.0221)
Avg. Rank	2.7500	2.6250	3.0625	5.2500	5.0625	4.3125	4.8750
Win count	3	5	4	1	1	1	1

using both NN and NN_d are the largest among all the datasets; $N1$ values obtained using both NN and NN_d are also high for the Oil dataset. In addition to this, these two datasets are the only datasets whose $N1$ values with NN_d are greater than those with NN except for Segment, which shows very low $N1$ values. If the proportion of samples likely to generate noise during the sampling procedure by SMOTE is high, some of them are given the same sampling probability value as other

TABLE 11. Average AUROC values for the original datasets: Comparison between NASBagging and SMOTEBagging.

Data	NASBagging($R1$)		NASBagging($R2$)		SMOTEBagging
	$M1$	$M2$	$M1$	$M2$	
Abalone	0.8337 (0.0027)	**0.8342** (0.0030)	0.8325 (0.0037)	0.8302 (0.0042)	0.8299 (0.0053)
Ecoli	0.9992 (0.0005)	0.9995 (0.0005)	**0.9996** (0.0005)	0.9996 (0.0005)	0.9994 (0.0006)
Hepatitis	0.8201 (0.0328)	0.8264 (0.0322)	0.8165 (0.0277)	0.8260 (0.0298)	**0.8264** (0.0262)
Ionosphere	0.9657 (0.0031)	**0.9661** (0.0035)	0.9606 (0.0034)	0.9617 (0.0061)	0.9642 (0.0035)
Isolet	**0.9764** (0.0018)	0.9751 (0.0017)	0.9731 (0.0020)	0.9712 (0.0021)	0.9729 (0.0022)
KC1	0.8003 (0.0029)	0.8052 (0.0053)	0.7790 (0.0042)	**0.8125** (0.0039)	0.7799 (0.0060)
Letter Img	0.9964 (0.0011)	**0.9978** (0.0009)	0.9962 (0.0007)	0.9977 (0.0009)	0.9961 (0.0014)
Mammography	0.8736 (0.0099)	0.8792 (0.0069)	0.7888 (0.0079)	**0.9244** (0.0089)	0.7881 (0.0053)
Oil	0.8483 (0.0190)	**0.8619** (0.0192)	0.8400 (0.0223)	0.8479 (0.0221)	0.8432 (0.0145)
Optical digits	0.9895 (0.0022)	0.9876 (0.0014)	0.9893 (0.0016)	0.9876 (0.0019)	**0.9895** (0.0007)
PC1	0.8521 (0.0144)	0.8532 (0.0117)	0.8527 (0.0150)	0.8512 (0.0146)	**0.8545** (0.0127)
Satimage	**0.9466** (0.0016)	0.9449 (0.0027)	0.9446 (0.0023)	0.9440 (0.0029)	0.9439 (0.0028)
Spectrometer	0.9270 (0.0136)	**0.9363** (0.0128)	0.9261 (0.0120)	0.9346 (0.0126)	0.9246 (0.0149)
US crime	0.8783 (0.0078)	**0.8795** (0.0065)	0.8728 (0.0060)	0.8718 (0.0107)	0.8698 (0.0063)
Wine quality	0.8096 (0.0100)	**0.8207** (0.0087)	0.8043 (0.0115)	0.8061 (0.0075)	0.8008 (0.0104)
Yeast	0.8812 (0.0138)	**0.8845** (0.0163)	0.8627 (0.0206)	0.8651 (0.0186)	0.8523 (0.0128)
Avg. Rank	2.5625	1.6875	3.6875	3.1875	3.8750
Win count	2	8	1	2	3

samples and their sampling probability is higher than that in SMOTEBoost, which may degrade the performance of NASBoost for these datasets.

Figure 2 shows how many times synthetic samples are generated from each sample at each iteration; these values are obtained using varying methods for the dataset, KC1. KC1 is selected because the diversity in the difficulty level of the samples, calculated using $N1$, is high. Twenty-five samples in the minority class were randomly selected to have various N1 values in the subset. In each subfigure, blue-colored columns of pixels correspond to each iteration, and each row of pixels corresponds to each sample. The different blue colors of the pixels from the 1st to 50th columns are related to the number of synthetic samples from the selected samples. The last three columns, represented by different red colors, show $N1$, $R1$, and $R2$ values of the selected samples. The blue and red color maps are located at the right and bottom, respectively. In this figure, B2-SMOTEBoost is omitted because no difference exists between B1-SMOTEBoost and B2-SMOTEBoost in terms of which sample is selected to generate a synthetic sample.

As shown in Figure 2, SMOTEBoost shows the smallest variation in the number of synthetic samples derived from each sample, whereas WOTBoost and RAMOBoost show high variance. In particular, WOTBoost generates many synthetic samples from samples that were not properly classified in the previous iteration. Samples in which WOTBoost imposes a high sampling probability have high values of $R1$ and $R2$. Consequently, the possibility of generating noise samples increases, which may be the main reason for the degradation in the classification performance of WOTBoost. NASBoost shows a greater deviation than SMOTEBoost in the number of synthetic samples derived from each sample, whereas it presents less deviation than RAMOBoost and WOTBoost. Instead of excluding the samples at the resampling step, SMOTE can be applied to samples with relatively low risk to generate noise among samples with high difficulty levels in classification, considering that samples with similar $N1$ values can have different levels of $R1$ or $R2$ values. Hence, NASBoost can maintain higher diversity among the training sets for the base classifiers than Borderline-SMOTEBoost.

Tables 9 and 10 compare the M2 models of NASBoost with other boosting algorithms for the noisy datasets with respect to AUROC and AUPR, respectively. The superiority of NASBoost is greater for the noisy datasets than for the original datasets, especially in terms of AUROC. Moreover, SMOTEBoost, which is not the best method among the comparison methods for the original datasets, becomes the best method among the comparison methods for the noisy datasets, regardless of the evaluation measures. When the proportion of noise samples in the minority class increases, the ratio of the generation of new samples from non-noise samples may decrease for RAMOBoost and WOTBoost. In the case of BorderlineSMOTEBoost, the ratios of samples not excluded in sampling may decrease when the proportion of noise samples in the minority class is high.

TABLE 12. Average AUPR values for the original datasets: Comparison between NASBagging and SMOTEBagging.

Data	NASBagging($R1$)		NASBagging($R2$)		SMOTEBagging
	$M1$	$M2$	$M1$	$M2$	
Abalone	0.2883 (0.2883)	**0.2909** (0.2859)	0.2859 (0.2881)	0.2886 (0.2831)	0.2844 (0.0073)
Ecoli	0.9932 (1.0000)	0.9960 (1.0000)	**0.9968** (1.0000)	0.9968 (1.0000)	0.9964 (0.0000)
Hepatitis	0.6337 (0.9932)	**0.6646** (0.9968)	0.6440 (0.9945)	0.6562 (0.9943)	0.6415 (0.0038)
Ionosphere	0.9344 (0.6337)	0.9460 (0.6440)	0.9294 (0.6491)	**0.9471** (0.6518)	0.9338 (0.0420)
Isolet	**0.8034** (0.9344)	0.7938 (0.9294)	0.7811 (0.9377)	0.7642 (0.9332)	0.7791 (0.0097)
KC1	0.4460 (0.8034)	0.4463 (0.7811)	0.3819 (0.7927)	**0.4567** (0.7635)	0.3873 (0.0125)
Letter Img	0.9754 (1.0000)	0.9800 (1.0000)	0.9772 (1.0000)	**0.9809** (1.0000)	0.9777 (0.0000)
Mammography	0.6210 (0.4460)	**0.6745** (0.3819)	0.4279 (0.4372)	0.6634 (0.4238)	0.4296 (0.0101)
Oil	0.3604 (0.9754)	0.3613 (0.9772)	**0.3715** (0.9785)	0.3613 (0.9795)	0.3607 (0.0031)
Optical digits	0.9508 (0.6210)	0.9452 (0.4279)	**0.9509** (0.6073)	0.9444 (0.5841)	0.9488 (0.0218)
PC1	0.3908 (0.3604)	0.3930 (0.3715)	**0.3962** (0.3551)	0.3942 (0.3561)	0.3939 (0.0226)
Satimage	0.7238 (0.9508)	**0.7278** (0.9509)	0.7254 (0.9440)	0.7264 (0.9438)	0.7248 (0.0034)
Spectrometer	0.7070 (0.3908)	0.7439 (0.3962)	0.7013 (0.3524)	**0.7454** (0.3675)	0.7002 (0.0323)
US crime	**0.4293** (0.7238)	0.4231 (0.7254)	0.4137 (0.7227)	0.4097 (0.7239)	0.4121 (0.0075)
Wine quality	0.3170 (1.0000)	**0.3173** (1.0000)	0.3090 (1.0000)	0.3164 (1.0000)	0.3112 (0.0000)
Yeast	0.3551 (0.7070)	**0.3718** (0.7013)	0.3427 (0.7332)	0.3558 (0.7364)	0.3491 (0.0224)
Avg. Rank	3.3750	2.0000	3.3125	2.4375	3.8750
Win count	2	6	4	4	0

TABLE 13. Average AUROC values for the noisy datasets: Comparison between NASBagging and SMOTEBagging.

Data	NASBagging($R1$)		NASBagging($R2$)		SMOTEBagging
	$M1$	$M2$	$M1$	$M2$	
Abalone	0.7802 (0.0065)	**0.7820** (0.0077)	0.7750 (0.0050)	0.7755 (0.0065)	0.7728 (0.0071)
Ecoli	**0.9635** (0.0115)	0.9590 (0.0097)	0.9593 (0.0093)	0.9607 (0.0110)	0.9617 (0.0111)
Hepatitis	**0.8102** (0.0318)	0.8100 (0.0312)	0.8046 (0.0340)	0.8010 (0.0361)	0.7953 (0.0450)
Ionosphere	**0.9132** (0.0052)	0.9108 (0.0067)	0.9099 (0.0070)	0.9060 (0.0085)	0.9101 (0.0099)
Isolet	**0.9302** (0.0021)	0.9263 (0.0022)	0.9147 (0.0024)	0.9073 (0.0032)	0.9137 (0.0034)
KC1	0.7415 (0.0082)	0.7508 (0.0076)	0.7047 (0.0074)	**0.7699** (0.0067)	0.7046 (0.0078)
Letter Img	0.9539 (0.0020)	**0.9556** (0.0019)	0.9537 (0.0022)	0.9541 (0.0016)	0.9528 (0.0013)
Mammography	0.7884 (0.0058)	0.7870 (0.0084)	0.7607 (0.0054)	**0.8775** (0.0052)	0.7634 (0.0054)
Oil	**0.8281** (0.0315)	0.8273 (0.0303)	0.8199 (0.0362)	0.8214 (0.0353)	0.8216 (0.0249)
Optical digits	0.9425 (0.0026)	**0.9437** (0.0038)	0.9427 (0.0037)	0.9414 (0.0043)	0.9433 (0.0040)
PC1	0.8096 (0.0128)	0.8122 (0.0140)	0.8133 (0.0118)	0.8141 (0.0143)	**0.8178** (0.0142)
Satimage	**0.9050** (0.0039)	0.9027 (0.0045)	0.9003 (0.0043)	0.8983 (0.0040)	0.8990 (0.0052)
Spectrometer	0.8953 (0.0136)	**0.8990** (0.0143)	0.8807 (0.0132)	0.8845 (0.0165)	0.8871 (0.0129)
US crime	0.8242 (0.0065)	**0.8280** (0.0097)	0.8136 (0.0069)	0.8113 (0.0083)	0.8095 (0.0092)
Wine quality	0.7907 (0.0093)	**0.7955** (0.0096)	0.7859 (0.0113)	0.7891 (0.0093)	0.7851 (0.0080)
Yeast	0.8122 (0.0132)	**0.8251** (0.0146)	0.8058 (0.0177)	0.8038 (0.0170)	0.7971 (0.0198)
Avg. Rank	2.0625	1.9375	3.7500	3.4375	3.8125
Win count	6	7	0	2	1

2) BAGGING

Tables 11 and 12 compare models of NASBagging and SMOTEBagging for the original datasets based on AUROC and AUPR, respectively. The average ranks and win counts in the two tables consider all the models because only two variations exist in NASBagging (M1 and M2). In addition,

TABLE 14. Average AUPR values for the noisy datasets: Comparison between NASBagging and SMOTEBagging.

Data	NASBagging($R1$)		NASBagging($R2$)		SMOTEBagging
	$M1$	$M2$	$M1$	$M2$	
Abalone	0.2704 (0.2715)	**0.2715** (0.2670)	0.2610 (0.2704)	0.2670 (0.2627)	0.2635 (0.0064)
Ecoli	0.9022 (0.9319)	0.9069 (0.9313)	0.8972 (0.9316)	0.9084 (0.9275)	**0.9121** (0.0147)
Hepatitis	**0.6639** (0.9069)	0.6629 (0.9084)	0.6433 (0.8948)	0.6560 (0.8993)	0.6396 (0.0103)
Ionosphere	0.8863 (0.6629)	**0.9008** (0.6560)	0.8795 (0.6556)	0.8939 (0.6470)	0.8720 (0.0315)
Isolet	0.7116 (0.9008)	**0.7441** (0.8939)	0.6465 (0.8883)	0.6802 (0.8820)	0.6672 (0.0117)
KC1	0.3724 (0.7441)	0.3893 (0.6802)	0.3028 (0.7144)	**0.4296** (0.6472)	0.4082 (0.0109)
Letter Img	**0.8788** (0.9430)	**0.8788** (0.9434)	0.8722 (0.9433)	0.8765 (0.9437)	0.0461 (0.0010)
Mammography	0.3975 (0.3893)	0.4585 (0.3222)	0.3171 (0.3801)	**0.5837** (0.3650)	0.0283 (0.0076)
Oil	0.3547 (0.8754)	**0.3678** (0.8765)	0.3544 (0.8785)	0.3657 (0.8731)	0.3646 (0.0004)
Optical digits	0.8800 (0.4585)	0.8855 (0.3890)	0.8695 (0.4178)	**0.8855** (0.4510)	0.8838 (0.0011)
PC1	0.3715 (0.3678)	0.3799 (0.3657)	0.3822 (0.3562)	**0.3842** (0.3575)	0.2827 (0.0306)
Satimage	0.6739 (0.8855)	**0.6814** (0.8855)	0.6775 (0.8794)	0.6779 (0.8700)	0.6766 (0.0034)
Spectrometer	0.6854 (0.3799)	**0.6950** (0.3842)	0.6451 (0.3715)	0.6576 (0.3806)	0.6361 (0.0327)
US crime	0.3815 (0.6814)	**0.3898** (0.6779)	0.3640 (0.6762)	0.3789 (0.6767)	0.3803 (0.0038)
Wine quality	0.2800 (0.9346)	**0.2857** (0.9350)	0.2785 (0.9299)	0.2783 (0.9276)	0.0450 (0.0015)
Yeast	0.2920 (0.6835)	**0.3091** (0.6576)	0.2929 (0.6885)	0.2952 (0.6519)	0.0864 (0.0230)
Avg. Rank	2.9375	1.5625	4.1250	2.3125	4.0000
Win count	2	10	0	4	1

the experimental results for the noisy datasets that compare the models of NASBagging and SMOTEBagging are displayed in Tables 13 and 14.

Similar to NASBoost, M2 showed the better results than M1. Whether it uses M1 and M2, NASBagging outperforms SMOTEBagging for the original and noisy datasets; but, based on both evaluation metrics, the enhancement obtained by using NASBagging is greater for the noisy datasets than for the original datasets. NASBagging is superior to SMOTEBagging even for Oil and Wine quality datasets unlike NASBoost. In NASBagging and SMOTEBagging, some new samples added into training sets consist of replication of existing samples by random selection. Since Oil and Wine quality include lots of the noise minority class samples, the noise samples can become multiple in the training sets. This might be the reason why NASBagging, which prevents SMOTE from not using noise in resampling, can perform better than SMOTEBagging.

3) SUMMARY

To verify whether NASBoost outperforms other comparison methods, some statistical tests such as the pairwise t-test and Nemenyi pairwise test [51] were applied to the experimental results. First, Table 15 provides the number of datasets for which the proposed method shows significantly better performance compared to other comparison methods using AUROC and AUPR, at a significance level of 0.05.[4] In this table, NAS represents NASBoost and the type of model used for the t-test is M2. Regardless of the comparison methods used, NASBoost with $R1$ and $R2$ is significantly better for least 10 datasets for the original datasets and 13 datasets for the noisy datasets, which also supports the idea that NASBoost works well with noisy data.

TABLE 15. Summary of the pairwise $t-$test for NASBoost.

Comparison methods	Original		Noisy	
	NAS($R1$)	NAS($R2$)	NAS($R1$)	NAS($R2$)
AUROC				
SMOTEBoost	13	13	13	13
B1-SMOTE	15	15	16	16
B2-SMOTE	13	13	16	16
RAMO	11	11	15	15
WOT	11	11	16	16
AUPR				
SMOTEBoost	10	10	13	13
B1-SMOTE	15	15	16	16
B2-SMOTE	13	13	16	16
RAMO	11	11	13	13
WOT	12	12	14	14

Next, Table 17 summarizes the p-values of the Nemenyi pairwise test for pairs of NASBoost and individual comparison methods. The Nemenyi pairwise test was not applied to the experimental results based on bagging, because there was only one comparison method, SMOTEBagging. For both types of datasets, NASBoost is significantly better than B1-SMOTEBoost, B2-SMOTEBoost, and WOTBoost using AUROC and AUPR, at a significance level of 0.05. However, based on the Nemenyi test, it can be concluded that NASBoost is significantly better than SMOTEBoost and RAMOBoost. This is because, compared to other comparison methods, SMOTEBoost and RAMOBoost show high rankings for many datasets.

[4]The p-values of the t-test are provided in Section S2 of the supplementary material.

TABLE 16. Summary of the pairwise t−test for NASBagging compared with SMOTEBagging.

Metric	Original		Noisy	
	NAS($R1$)	NAS($R2$)	NAS($R1$)	NAS($R2$)
AUROC	14	14	15	15
AUPR	15	15	9	9

TABLE 17. p-values of the Nemenyi pairwise test for NASBoost.

Comparison methods	Original		Noisy	
	NAS($R1$)	NAS($R2$)	NAS($R1$)	NAS($R2$)
AUROC				
SMOTEBoost	0.299	0.408	0.531	0.935
B1-SMOTE	0.017	0.014	0.002	0.009
B2-SMOTE	0.040	0.046	0.000	0.000
RAMO	0.717	0.717	0.003	0.170
WOT	0.009	0.012	0.000	0.000
AUPR				
SMOTEBoost	0.993	0.986	1.000	1.000
B1-SMOTE	0.002	0.001	0.012	0.012
B2-SMOTE	0.017	0.010	0.040	0.030
RAMO	0.170	0.207	0.351	0.250
WOT	0.003	0.003	0.049	0.030

However, it can be concluded that the proposed method is effective for imbalanced data because it shows better performance for most of the datasets compared with SMOTEBoost and RAMOBoost according to the paired t-test.

V. CONCLUSION

For imbalanced datasets, the classification performance is linked to not only the imbalance ratio but also other factors related to data distributions, such as the presence of borderline and noise samples. In particular, the presence of noise can hinder the performance of ensemble models. This condition is severe in boosting algorithms. However, most SMOTE-based sampling methods combined with ensemble algorithms determine the borderline and noise samples based on the class distribution within the kNN. Even in the case of borderline samples, new synthetic samples can be in the region of the majority class. Therefore, whether to apply sampling to this sample should be decided in consideration of the process of generating a synthetic sample in SMOTE.

To address this issue, this study proposes a new modification of SMOTE for ensemble approaches combined with resampling to deal with class imbalance. The proposed sampling method is designed to avoid selecting samples in the minority class with a high probability of introducing noise to the region of the majority class while maintaining diversity among training sets. The main contributions of this study are summarized as follows:

- New metrics to precisely measure the degree of risk with which a noise sample is synthesized, $R1$ and $R2$ are proposed using the classification complexity measures.
- A new oversampling procedure using $R1$ and $R2$ is designed to effectively suppress noise generation by resampling while maintaining diversity in the training sets.
- The performances of NASBoost and NASBagging according to methods to reflect the risk of noise generation in resampling are examined in detail.
- The superiority of NASBoost and NASbagging over comparison methods has been extensively validated on 16 datasets for AUROC and AUPR, which are popular evaluation metrics for imbalanced data.

In conclusion, NASBoost and NASBagging work well even for datasets with a considerable number of noise samples, as intended. The experiments also show that considering the prediction results of previous classifiers even at the sampling step can degrade the performance of ensemble models in the case of boosting, which might be due to excessive overfitting on noise samples. In other words, a strategy to maintain diversity among training sets of NASBoost and NASBagging as much as possible is effective in dealing with class imbalance. Moreover, as in the comparison results between $R1$ and $R2$, it can be seen that it is better to avoid noise generation in resampling by considering the class distribution of neighbor samples rather than considering the average distance to different classes.

The results for Oil and Wine quality datasets show that the performance of NASBoost may degrade when the proportion of minority class samples that are likely to generate noise is high. Therefore, it is necessary to study a better method for calculating the sampling probability from $R1$ and $R2$ considering the proportion of such samples. In addition, other methods to measure the risk of introducing noise samples using SMOTE will be studied.

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